

Last class

- Recap of linear spaces
- " " metric spaces
- $\mathcal{F}(S, \mathbb{K})$, $S \neq \emptyset$.
- $C(\Omega)$
- Other examples.

Normed linear Space.

$X \neq \emptyset$ linear space over $\mathbb{K}(\mathbb{R}/\mathbb{C})$.

$$\|\cdot\| : X \rightarrow \mathbb{R}.$$

$$N1) \quad \begin{array}{l} \|\alpha\| \geq 0 \quad \forall \alpha \in X \\ \|\alpha\| = 0 \iff \alpha = 0 \end{array} \quad \left. \vphantom{\begin{array}{l} \|\alpha\| \geq 0 \\ \|\alpha\| = 0 \iff \alpha = 0 \end{array}} \right\} \text{positivity}$$

$$N2) \quad \|\alpha \alpha\| = |\alpha| \|\alpha\|, \quad \begin{array}{l} \forall \alpha \in X \\ \alpha \in \mathbb{K} \end{array} \quad \left. \vphantom{\begin{array}{l} \forall \alpha \in X \\ \alpha \in \mathbb{K} \end{array}} \right\} \text{homogeneity}$$

$$N3) \quad \|\alpha + \beta\| \leq \|\alpha\| + \|\beta\| \quad \left. \vphantom{\|\alpha + \beta\|} \right\} \text{triangle ineq.}$$

Then $\|\cdot\|$ is called a norm and $(X, \|\cdot\|)$ is called a normed linear space.

Remark:

Given a nls $(X, \|\cdot\|)$
define $d(x, y) = \|x - y\| \quad \forall x, y \in X$.
 $d \rightarrow$ satisfies all the properties of
metric.

and (X, d) is a metric space.

'd' induced from the norm.

Seminorm.

N(2) automatically gives

Suppose $x = 0$

$$\begin{aligned}\|x\| &= \|0\| = \|0 \cdot 0\| \\ &= |0| \|0\| = 0.\end{aligned}$$

That gives $\Rightarrow x = 0 \Rightarrow \|x\| = 0$.

- N_1 satisfies partially $\left. \begin{array}{l} \|x\| \geq 0 \\ \| \cdot \| \rightarrow \text{seminorm.} \end{array} \right\}$
- N_2 • N_3

Ex.

$(X, \| \cdot \|)$ is a nls.

$$d(x, y) = \|x - y\|$$

- Translation invariance

$$d(x+u, y+u) = d(x, y)$$

- Homogeneity

$$d(\lambda x, \lambda y) = |\lambda| d(x, y)$$

Remark:

X s.t d is a metric on a lsp.
 d is either not

T.I or not Homogenous then
such a metric d cannot
be induced from norm.

Ex.
$$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}$$
$$X = \mathbb{R}. \quad x, y \in X.$$

Qn X is a linear space
sp. We have a metric
' d ' s.t d satisfies

i) T I.

ii) Homogeneity.

Is it true that $\exists$ a norm
on X ~~set~~ which induces
 d ?

Ans: Yes!!

Define $\|x\| = d(0, x)$

- Verify $\|\cdot\|$ satisfies all properties of norm.

• If ρ is a metric
induced by $\|\cdot\|$
then $\rho = d$. $\textcircled{*}$

$$\begin{aligned}\rho(x, y) &= \|x - y\| = d(0, x - y) \\ &= d(0 + y, x - y + y) \\ &= d(y, x) \\ &= d(x, y) \\ &\quad \underline{\underline{\forall x, y \in X.}}\end{aligned}$$

Convergence

Let $(X, \|\cdot\|)$ be a nls
and (x_n) converges to $x \in X$
if $\forall \epsilon > 0 \quad \exists N \in \mathbb{N}_\epsilon$ s.t

$$\|x_n - x\| < \epsilon \quad \forall n \geq N_\epsilon$$

Cauchy Seq

A seq (x_n) in a nls
 $(X, \|\cdot\|)$ is said to
be Cauchy seq if $\forall \epsilon > 0$

$\exists N_\epsilon \in \mathbb{N}$ s.t

$$\|x_n - x_m\| < \epsilon \quad \forall n, m \geq N_\epsilon.$$

Banach Space.

If every Cauchy seq in
a NLS is convergent then
it is a Banach Space
(complete NLS).

Recall

Lipschitz fn.

(X, d)

(Y, ρ)

$$f: X \rightarrow Y.$$

f is Lipschitz iff $\exists K > 0$
s.t

$$\rho(f(x), f(y)) \leq K d(x, y).$$

• Lipschitz fm $\Rightarrow$ v.c.

Prop. In a nls $(X, \|\cdot\|)$
we have

$$* \quad |\|x\| - \|y\|| \leq \|x - y\|$$

$\forall x, y \in X.$

$$f = \|\cdot\| : X \rightarrow (\mathbb{R}, |\cdot|)$$

$$\rightarrow |f(x) - f(y)| \leq \|x - y\| = d(x, y)$$

$$|f(x) - f(y)| \leq 1 \cdot d(x, y)$$

$$\Rightarrow \|\cdot\| \text{ is v.c.}$$

Proof:

$$\|x\| = \|x - y + y\|$$

$$\leq \|x - y\| + \|y\|$$

$$\|x\| - \|y\| \leq \|x - y\| \quad \text{--- (1)}$$

Interchange the role of
 x and y then we get

$$\|y\| - \|x\| \leq \|x - y\| \quad (2)$$

① Δ ②

$$|\|x\| - \|y\|| \leq \|x - y\|$$

Prop. Let $(X, \|\cdot\|)$ is a nls.

Let $(x_n), (y_n)$ in X be

s.t $x_n \rightarrow x$ in X

$y_n \rightarrow y$ in X .

Let (λ_n) in $\mathbb{K}$ s.t
 $\lambda_n \rightarrow \lambda$ as $n \rightarrow \infty$ in $\mathbb{K}$.

Then $\lambda_n x_n \rightarrow \lambda x$
 $x_n + y_n \rightarrow x + y$.

Skip.

Remark.

$(X, \|\cdot\|)$

$+$: $X \times X \rightarrow X$.

$\bullet$: $\mathbb{K} \times X \rightarrow X$.

Recall.

a fn $f: X \rightarrow Y$

is continuous $\Leftrightarrow$

$$\forall x_n \rightarrow x$$

$$\Rightarrow f(x_n) \rightarrow f(x) \text{ as } n \rightarrow \infty.$$

X x X.

:

$$(x_n, y_n) \rightarrow (x, y) \text{ as } n \rightarrow \infty$$

$\Leftrightarrow$

$$\left. \begin{array}{l} x_n \rightarrow x \\ \& y_n \rightarrow y \end{array} \right\}.$$

' + '

$$(x_n, y_n) \rightarrow (x, y)$$

$$+ (x_n, y_n) = x_n + y_n \rightarrow + (x, y)$$

' + ' is continuous fn. on $X \times X$
 ' . ' is " " on $\mathbb{K} \times X$.

Lemma. $(X, \|\cdot\|)$ a nls.

X_0 be a subspace of X .

Then $\overline{X_0}$ is a closed subspace of X .

Pr. Hint:

$$x, y \in \overline{X_0}$$

$$\& \lambda \in \mathbb{K}.$$

$$\text{Then } \lambda x + y \in \overline{X_0} \quad (?)$$

Examples.

$$\text{i) } \mathbb{K} = \mathbb{R} / \mathbb{C}.$$

$$\forall x \in \mathbb{K}, \quad \|x\| = |x|.$$

Remark. X is a v.s over $\mathbb{K}$.

$$\|x\|_{\alpha} = c \|x\| \quad \forall x \in X.$$

In particular $c > 0$.
 $\mathbb{K} = \mathbb{R} / \mathbb{C}.$

Any norm will be a
const multiple of l.l.

$$\begin{aligned} \|x\| &= \|x \cdot 1\| = |x| \|1\| \\ &= c |x| \end{aligned}$$

$$\textcircled{2} \quad \mathbb{K}^n \cong \mathcal{F}(S, \mathbb{K});$$

$$S = \{1, 2, 3, \dots, n\}.$$

For each

$$x = (x(1), x(2), \dots, x(n)) \\ \in \mathbb{K}^n.$$

Define,

$$\|x\| = \sum_{i=1}^n \underline{|x(i)|}.$$

Then $\|\cdot\|$ is a norm on $\mathbb{K}^n$.

For triangle ineq observe

$$|x(i) + y(i)| \leq |x(i)| + |y(i)|$$

$$x = (x(1), \dots, x(n))$$

$$y = (y(1) \dots y(n)) \in \mathbb{K}^n.$$

3. For $x \in \mathbb{K}^n$,

$$\|x\|_{\infty} = \max_{1 \leq i \leq n} |x(i)|$$

$\|\cdot\|_{\infty}$ is a norm.

Ineq.

• Cauchy-Schwarz $\times$

• Hölder's $\times$

• Minkowski's ineq.

$$\| \cdot \|_p \quad \| \cdot \|_2 \quad \mathbb{K}^n$$